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The role of hypersalinity in the persistence of the Texas 'brown tide' in the Laguna Madre

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Abstract. A brown tide bloom of the alga *Aureoumbra lagunensis* was present without interruption in the Laguna Madre of Texas from January 1990 through October 1997. This is the longest continual phytoplankton bloom of which we are aware. Although the factors leading to the initiation of this bloom have been well documented, the factors contributing to its persistence are still being investigated. Two physical characteristics of the Laguna Madre may play an important role: the long turnover time for waters in this coastal lagoon (~1 year) and the hypersaline conditions that usually exist (40–60 PSU) due to evaporation exceeding precipitation. In this study, we examined the effects of salinity on the growth rates of the brown tide alga and on the growth of one of its protozoan grazers. Historical data from before the onset of the brown tide provide evidence for the suppression of microzooplankton populations and mesozooplankton growth caused by hypersalinity. The brown tide alga will grow in a remarkably wide range of salinities ranging from 10 to 90 PSU. Maximum growth rates are achieved at salinities ranging from 20 to 60 PSU. One common grazer on the brown tide alga, the heterotrophic dinoflagellate *Oxyrrhis marina*, was found to grow more slowly under hypersaline conditions. The normally hypersaline conditions of the Laguna Madre may, therefore, favor the brown tide alga over other phytoplankton species that do not grow well under hypersaline conditions, and also suppress the growth and feeding rates of potential grazers.

Introduction

The Texas brown tide was a dense, persistent bloom of *Aureoumbra lagunensis*, a small (4–5 μm diameter) pelagophyte that had not been scientifically described prior to the outbreak of this bloom (DeYoe *et al.*, 1997). The bloom began in January 1990, following a period of extended drought which raised salinities in the Baffin Bay region of the Laguna Madre to ~60 PSU. A series of two extreme cold fronts in December 1989 caused water temperatures in the Laguna Madre to fall below freezing (0°C) for up to 36 h (Buskey *et al.*, 1997). This bloom persisted without interruption from January 1990 through October 1997, although the areal extent of the bloom has varied markedly on a seasonal and annual basis (Buskey *et al.*, 1996). Under bloom conditions, *A.lagunensis* was usually present at cell densities ranging from 0.5 to 5×10^6 cells ml^{-1} (Buskey *et al.*, 1996). At densities $>1 \times 10^6$ cells ml^{-1}, the waters had a brownish color, and water transparency declined markedly.

The Laguna Madre is a large (2.15×10^5 ha), shallow (average depth 1.2 m), often hypersaline coastal lagoon (Armstrong, 1987) that contains extensive seagrass beds (Quammen and Onuf, 1993). The reduced water clarity associated with this persistent bloom reduced the distribution of seagrasses in deeper waters due to the attenuation of light (Dunton, 1994; Onuf, 1996). The extended brown tide also had negative impacts on other parts of the Laguna Madre ecosystem, including reductions in the biomass and diversity of planktonic and benthic grazers on phytoplankton (Buskey and Stockwell, 1993; Montagna *et al.*, 1993; Buskey *et al.*, 1997).

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Although the factors contributing to the initiation of this bloom have been well described (Stockwell *et al.*, 1993; DeYoe and Suttle, 1994; Buskey *et al.*, 1997), the reasons for its unusual persistence remain unresolved. One factor that is likely to contribute to the persistence of this bloom is the long turnover times for the physically isolated waters of Laguna Madre. Since tidal exchanges are very low with the Gulf of Mexico and evaporation often exceeds precipitation in this system, turnover times are difficult to estimate, but it is thought that they exceed 1 year (Shormann, 1992). However, the small amounts of freshwater inflow and long turnover time for this system also make it difficult to understand the source of nutrients fueling the continued bloom. Since *A.lagunensis* is unable to use nitrate (DeYoe and Suttle, 1994), it may rely on ammonium released by the decomposition of seagrasses that were severely reduced in biomass during this bloom (Onuf, 1996) to supplement the limited nutrient supply from natural run-off.

Another important factor in the persistence of this bloom is the hypersaline conditions that often exist in the Laguna Madre. Extreme hypersalinity is hypothesized to contribute to the persistence of the bloom in two ways: it may adversely affect potential planktonic and benthic grazers on phytoplankton (Buskey *et al.*, 1997), and it may adversely affect other species of phytoplankton that might compete with *A.lagunensis* for resources. In this study, we examined the relationships between salinity and brown tide, microzooplankton, and mesozooplankton populations before and during the bloom. We also examined the effects of light intensity, temperature and salinity on the growth of *A.lagunensis*, and the effects of temperature and salinity on the growth rate of one of the more common grazers on the brown tide, the heterotrophic dinoflagellate *Oxyrrhis marina*.

Method

Replicate subsurface whole-water samples were collected at 14 stations in Baffin Bay and the upper Laguna Madre (Figure 1). Samples were preserved with ~10% acid Lugol's iodine. Brown tide cell densities were estimated based on replicate cell counts using a hemocytometer and a Wild M20 microscope for dense samples collected after January 1990. Microzooplankton (protozoa, rotifers, copepod nauplii) and lower density brown tide samples (August–December 1989) were enumerated from whole-water samples using the Utermöhl method where 5–20 ml of sample (depending on plankton density) were settled in a well slide and counted using an inverted microscope. The brown tide alga cannot be positively identified with light microscopy based on morphological characteristics alone. Cells counted as brown tide possessed their characteristic size and shape, but these counts may be subject to error, especially at low densities prior to the onset of the bloom. The recent development of an immunofluorescence assay for *A.lagunensis* (Lopez-Barreiro *et al.*, 1998) has allowed the positive identification of brown tide cells in archived, formalin-preserved samples.

Mesozooplankton samples were collected using 20-cm-diameter, 153 μm mesh plankton nets mounted on a bongo frame at four stations (6, 8, 9 and 15) in Laguna Madre (Figure 1). The nets were towed below the surface for 5 min, and the volume of water sampled was estimated using a General Oceanics flowmeter

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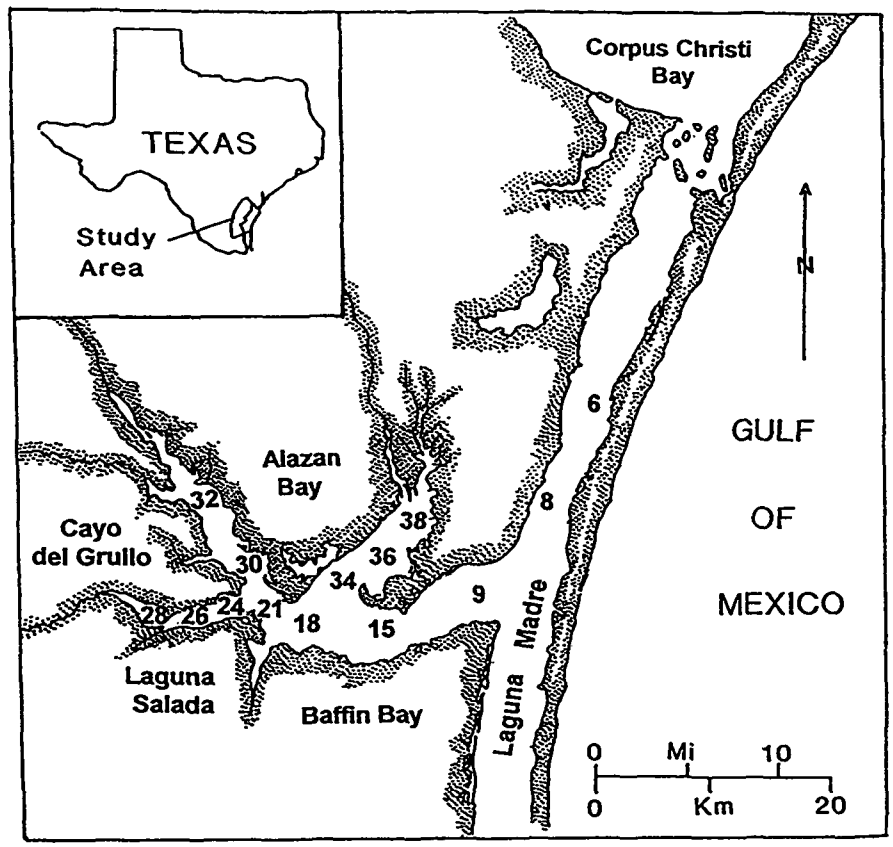

Fig. 1. Map of the upper Laguna Madre and Baffin Bay, Texas, showing the locations where whole-water samples were collected for counts of brown tide cell density and microzooplankton abundance (stations 18–36), and where zooplankton tows were taken (stations 6, 8, 9 and 15).

mounted in the mouth of each net. Samples were preserved in 5% buffered formaldehyde. Prosome lengths of 20 adult female *Acartia tonsa* from each sample were measured on a Wild M5 stereoscope, using an ocular micrometer.

Experiments were performed on clone TBA-2 of *A.lagunensis* isolated by F.Chen in March 1991 (DeYoe and Suttle, 1994). Light intensity was measured using a Li-Cor LI-250 light meter with a Li-Cor 193SA irradiance collector. Measurements were made by submerging the irradiance collector in culture media in a polycarbonate culture flask and obtaining a 15 s average of the light intensity. Care was taken to minimize shading during the measurement process. Light intensity was varied using layered, black plastic screening placed over culture flasks, and black cloth circles or white styrofoam circles placed beneath culture flasks. Fourteen light levels were achieved over two experiments ranging from 4 to 1650 μmol photons m^{-2} s^{-1}.

Aureoumbra lagunensis was cultured at 25°C on a 12:12 h light:dark (L:D) cycle under variable light intensity in 100 ml of modified f/2 media at 32 PSU in 250 ml

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polycarbonate culture flasks. Media modifications were as follows: K/20 levels of trace metals (Keller *et al.*, 1987) and NH_4Cl rather than $NaNO_3$. Triplicate cultures were inoculated at a target concentration of 2.5×10^4 cells ml^{-1} and sampled every 2 days at approximately the same time (~13:00–13:30 h) using sterile pipettes. Samples were preserved in 5% acid Lugol's solution and counted on the Z1 Coulter Counter within a few days of sampling. The mean value of four counts from each sample was plotted against time to monitor growth over the course of the experiment; the experiment was terminated at or near the end of logarithmic growth. The specific growth rate (μ) was determined by regressing the natural logarithm of cell concentration of cultures in logarithmic growth against time and taking the slope of the regression line. Each regression incorporated a minimum of four data points, each derived from a mean value of cell density from triplicate cultures.

Experiments were carried out to determine the effects of salinity on the growth rate of *A.lagunensis*. Cells were cultured at 25°C on a 12:12 h L:D cycle at ~300 μmol photons m^{-2} s^{-1} in 500 ml of modified f/2 in 600 ml polycarbonate flasks at 10, 15, 20, 30, 40, 50, 60, 70, 80 and 90 PSU salinity. High-salinity media (>30 PSU) were prepared in the laboratory by evaporating filtered sea water (0.2 μm porosity filter) collected from the Aransas ship channel. Low-salinity water was prepared by diluting filtered sea water with filtered distilled water. Triplicate cultures were inoculated at 7×10^4 cells ml^{-1}, and sampled every 2 days at approximately the same time (08:00–10:00 h). Various salinities were tested in three separate experiments. *Aureoumbra lagunensis* was grown at each of these salinities through at least three transfers before the start of these experiments to ensure that the cells had adapted to their new salinities.

In a second experiment, cells were cultured at 15°C on a 12:12 h L:D cycle at ~300 μmol photons m^{-2} s^{-1} in 500 ml of modified f/2 at 10, 15, 20, 30, 40, 50, 60, 70, 80 and 90 PSU salinity. Samples were collected at approximately the same time (08:00–10:00 h) every 3–4 days (due to slower growth).

The culture of *O.marina* used in this study was isolated from water collected in the Aransas ship channel near the University of Texas Marine Science Institute in Port Aransas, TX. Microzooplankton samples were collected with a 30-cm-diameter, 20 μm mesh plankton net allowed to stream with the tide. Aliquots of these plankton samples were diluted with filtered sea water and then incubated in 1 l polycarbonate bottles enriched with a mixture of cultured phytoplankton including *A.lagunensis*, *Dunaliella tertiolecta* and *Isochrysis galbana*. The enrichments were placed on a bottle roller rotating at ~2 r.p.m. and were incubated at 20°C at low light intensities for several days. When a protozoan species appeared to be growing well on the added phytoplankton species, individual cells were isolated under a stereomicroscope and brought into culture. *Oxyrrhis marina* cultures were maintained in 'ciliate media' (Gifford, 1985) in 50 ml tissue culture flasks held within a white PVC cylinder on a bottle roller under the same conditions as described for enrichments. Cultures were fed a mixture of *A.lagunensis* and *I.galbana*, and transferred to new media at weekly intervals.

Specific growth rates (μ; day^{-1}) of *O.marina* at various combinations of temperature and salinity were measured by adding 2–3 *O.marina* ml^{-1} to 200 ml

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of ciliate media containing *A.lagunensis* at a concentration equivalent to 1 mg carbon l^{-1} (~10^5 cells ml^{-1}). Previous studies have shown that this concentration of *A.lagunensis* supports rapid growth of *O.marina* (Buskey and Hyatt, 1995). Triplicate 10 ml samples preserved with 0.5 ml acid Lugol's solution were collected at the start of the experiment and daily at the same time for 4 days. Samples were settled in Utermöhl chambers and enumerated using an inverted microscope (Olympus IMT-2). Specific growth rates (μ; day^{-1}) were calculated from the linear portion of ln (organisms ml^{-1}) regressed against time (results were not used unless at least three of the five samples showed a linear relationship, since exponential growth cannot be confirmed for two data points).

Results

Before the onset of the brown tide, microzooplankton abundance, including mainly ciliates, tintinnids, rotifers and copepod nauplii, was strongly negatively correlated with salinity (Figure 2). At salinities below 60 PSU, microzooplankton abundance ranged from 20 to 100 organisms ml^{-1}. At salinities above 65 PSU, microzooplankton abundance was generally below 20 organisms ml^{-1}. After the onset of the bloom, microzooplankton abundance was generally below 20 organisms ml^{-1} throughout the range of salinities observed (45–75 PSU) and was only

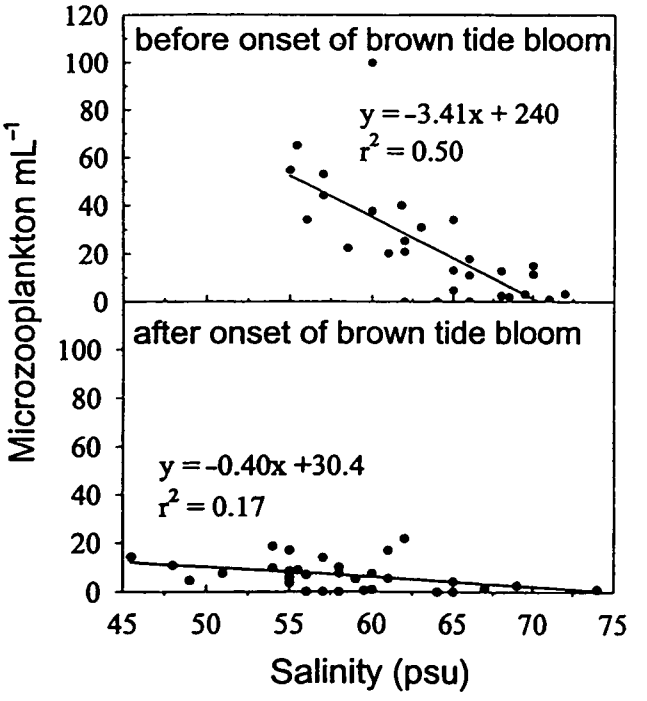

Fig. 2. Abundance of microzooplankton as a function of salinity in Baffin Bay before the onset of the brown tide bloom (**top**) and after the bloom began (**bottom**).

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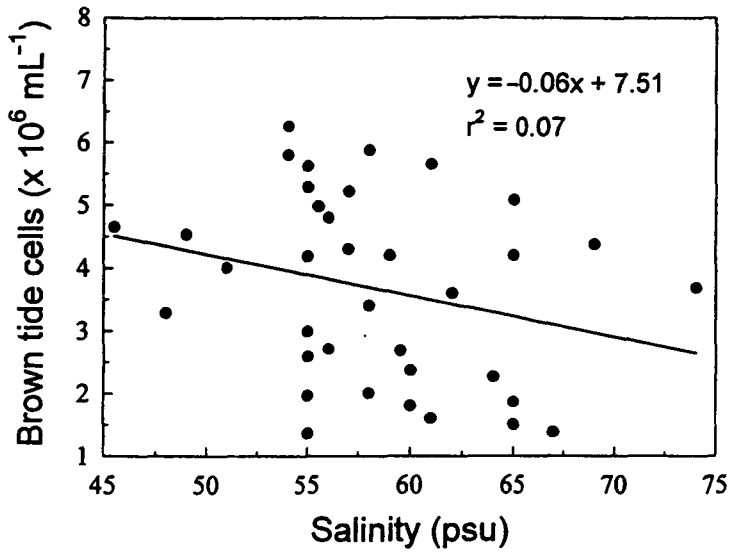

Fig. 3. Brown tide cell density as a function of salinity for stations in Baffin Bay after the onset of the bloom in January 1990.

very weakly negatively correlated to salinity (Figure 2). There was little correlation between brown tide cell density and salinity for samples collected in Baffin Bay after the initiation of the bloom (Figure 3).

The dominant mesozooplankter in the Laguna Madre, *Acartia tonsa*, also showed signs of being adversely affected by extremely high salinities before the onset of the brown tide bloom. Adult female *A.tonsa* showed a poor correlation between mean prosome length and salinities for individuals collected in Laguna Madre (stations 6 and 8; Figure 1) where salinities ranged between ~35 and 50 PSU (Figure 4). However, for copepods collected in Baffin Bay (stations 9 and 15), where salinities ranged from 50 to 65 PSU, there was a strong negative correlation between adult female prosome length and salinity.

The specific growth rate of *A.lagunensis* increased with light intensity until a maximum growth rate was achieved at ~100 μmol photons m^{-2} day^{-1}. No evidence was found for photoinhibition of growth at light intensities as high as 1650 μmol photons m^{-2} day^{-1} (Figure 5). Growth rates exceeding 0.5 day^{-1} were found for *A.lagunensis* at salinities ranging from 20 to 70 PSU at 25°C (Figure 6). *Aureoumbra lagunensis* showed a maximum specific growth rate of 0.61–0.63 day^{-1} at salinities ranging from 30 to 50 PSU at 25°C. Growth rates were below 0.4 day^{-1} at salinities of 15 and 80 PSU, and below 0.3 day^{-1} at salinities of 10 and 90 PSU. When grown at a temperature of 15°C, maximum growth rates of ~0.24 day^{-1} were found at salinities ranging from 30 to 50 PSU. Growth rates declined rapidly at salinities below 30 PSU or above 60 PSU at 15°C.

Specific growth rates for the heterotrophic dinoflagellate *O.marina* fed the brown tide alga *A.lagunensis* generally increased with increasing temperature at each of the three salinities tested (Figure 7). For each temperature, the fastest

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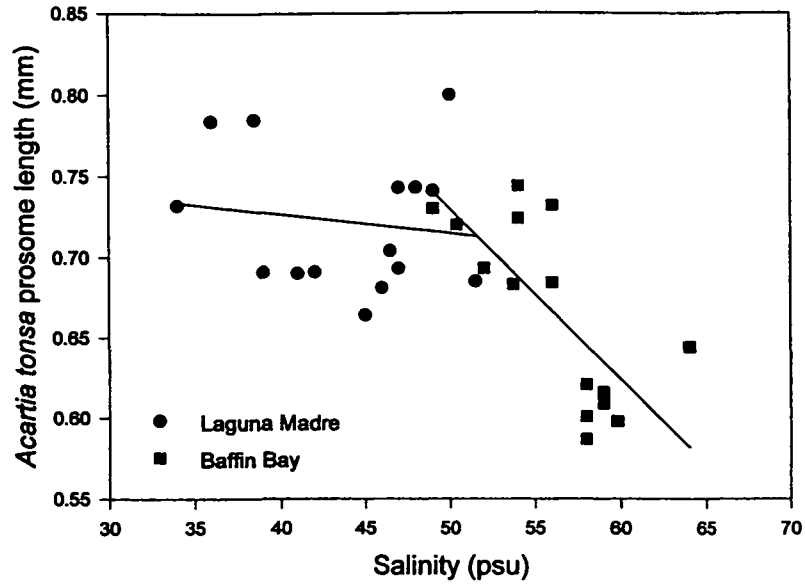

Fig. 4. Prosome length of *Acartia tonsa* adult females as a function of salinity for copepods collected at stations 6 and 8 in Laguna Madre (●) and at stations 9 and 15 in Baffin Bay (■).

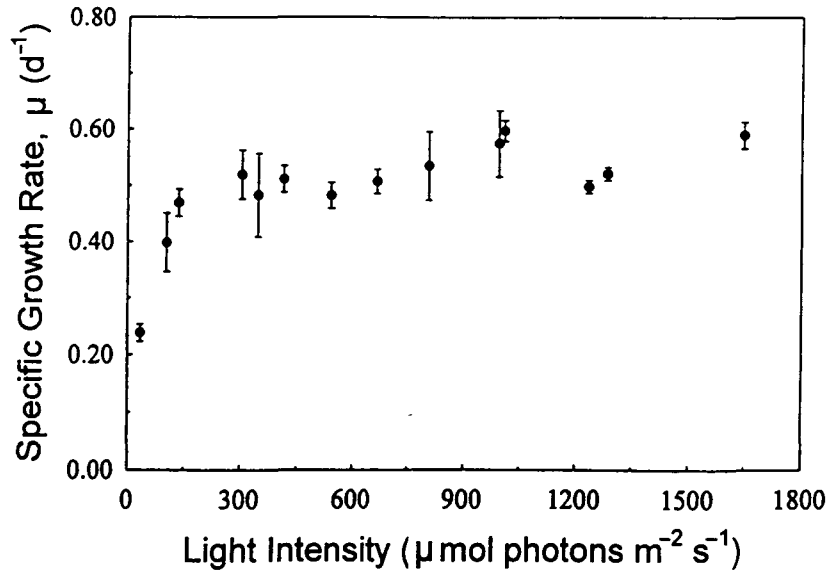

Fig. 5. Specific growth rate (day^{-1}) of *A.lagunensis* as a function of light intensity at 25°C and 32 PSU. Error bars indicate ± SE.

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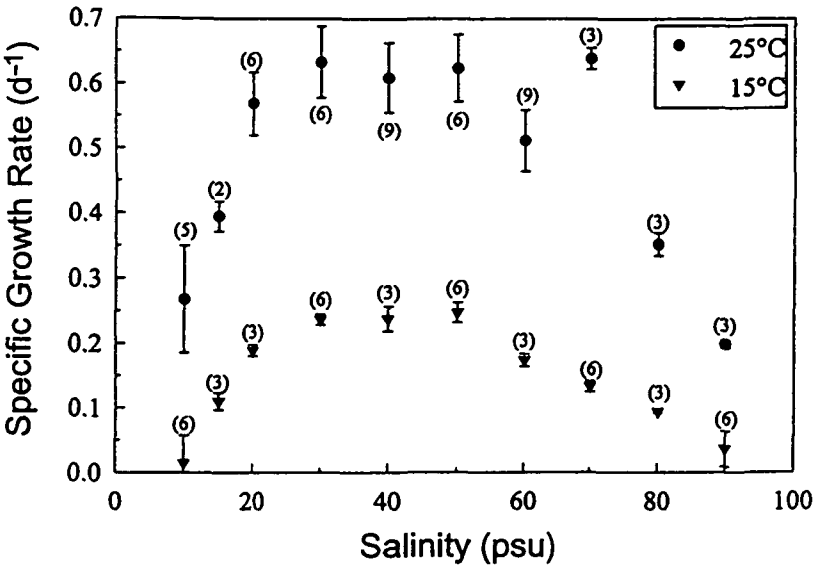

Fig. 6. Specific growth rate (day^{-1}) of *A.lagunensis* as a function of salinity at 25°C (●) or 15°C (▼) and a light intensity of 300 μmol photons m^{-2} s^{-1}. The numbers in parentheses refer to the number of replicate growth experiments at each salinity. Error bars indicate ± SE.

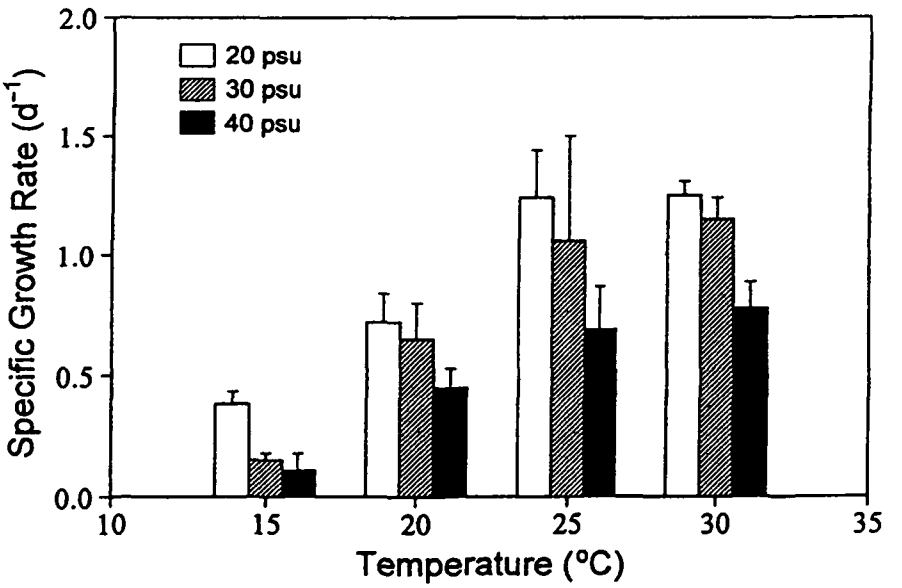

Fig. 7. Specific growth rate (day^{-1}) of *O.marina* as a function of temperature (15–30°C) and salinity (20–40 PSU). Error bars indicate ± SE.

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growth rates were achieved at a salinity of 20 PSU, with lower mean specific growth rates at 30 and 40 PSU. The maximum specific growth rate measured was 1.25 day^{-1} at 30°C and 20 PSU; the minimum specific growth rate was 0.11 day^{-1} at 15°C and 40 PSU. We were unable to maintain cultures at salinities ≥ 50 PSU.

Discussion

The role of physical and biological variables in the initiation and persistence of algal blooms is poorly understood. The persistent 'brown tide' bloom of *A.lagunensis* in the Laguna Madre of south Texas provides some unique insights into the dynamics of bloom events. Since the area where the bloom began had already been under study for 10 months before the initiation of the bloom, there is good information on the physical characteristics of the environment and its effects on the biological components of the system prior to the initiation of the bloom. The two most important physical factors in the initiation of the bloom appear to be the extremely hypersaline conditions that existed prior to the bloom and an unusually severe freeze that caused a massive die-off of fish and invertebrates (Buskey *et al.*, 1997); the decomposition of the dead organisms produced a large pulse of ammonia in the system that may have fueled the initial bloom (DeYoe and Suttle, 1994). These same conditions reduced the abundance of potential protozoan grazers on *A.lagunensis* prior to the bloom (Buskey *et al.*, 1997) and may have adversely affected potential phytoplankton competitors.

The tendency for hypersaline conditions in the Laguna Madre is a function of both the relatively slow exchange between this coastal lagoon and the Gulf of Mexico, and the low rainfall and high evaporation rates in this semi-arid region of the Texas coast. Evaporation generally exceeds precipitation by ~50 cm $year^{-1}$ (Brown *et al.*, 1977), and there are no rivers entering Baffin Bay, only intermittent streams. Exchange between Baffin Bay and the Gulf of Mexico is limited to the Port Mansfield channel ~80 km to the south and the Aransas Pass ~70 km toward the north. These limited routes for exchange lead to estimates of turnover rates in excess of 1 year for Baffin Bay (Shoremann, 1992).

Hypersaline environments are relatively uncommon worldwide and are limited in distribution, so the effects of hypersalinity on marine organisms has not been intensively studied. Most studies have been on species found in restricted extreme environments, such as inland salt pools and evaporation ponds (e.g. Friedman and Krumbein, 1985), or in the brine created during sea ice formation (e.g. Stoecker *et al.*, 1997). For example, ice algae are physiologically active at salinities as high as 100 PSU (Kottmeier and Sullivan, 1988). Most laboratory studies of the effects of salinity on common coastal marine organisms tend to concentrate on the effects of hyposalinity rather than hypersalinity. However, several studies have demonstrated higher mortalities for copepods (Moreira, 1975) and crab zoeae (Vernberg and Vernberg, 1975) under hypersaline conditions. Some phytoplankton species have rather narrow salinity ranges that allow for maximum growth, and may show reduced growth rates under hypersaline conditions (e.g. Kain and Fogg, 1958), while other species show maximal growth rates over a wider range of salinities (Kain and Fogg, 1960).

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There are relatively few studies of the plankton dynamics of hypersaline lagoons that maintain a direct connection to the sea, such as the Laguna Madre. Hedgepeth (1967) notes the general decrease in the number of species of marine organisms found in the Laguna Madre in salinities above 40 PSU. In an ecological study of the Alazon Bay region of Baffin Bay, Cornelius (1984) identified 90 different species of phytoplankton in waters of salinities ranging from 21 to 30 PSU, reducing to 55 species in waters of 31–40 PSU and only 15 species in waters of 41–50 PSU. Hypersalinity may also restrict the frequency of phytoplankton blooms. Hildebrand and King (1979) observed no phytoplankton blooms in waters above 40 PSU during a 6 year study of the upper Laguna Madre. Once a bloom organism such as *A.lagunensis* becomes established in a hypersaline lagoon, there may be relatively few potential competitors.

The Texas brown tide alga *A.lagunensis* has a remarkable tolerance to high salinities. It can sustain maximum growth rates at salinities up to 50 PSU, and growth greater than one half the maximum rate over a salinity range of 15–80 PSU. It also appears to be a warm water-adapted species with faster growth rates at 25°C than at 15°C. This agrees well with long-term field data on the abundance of *A.lagunensis* during the bloom. Typical winter water temperatures in the Laguna Madre are ~15°C, and *A.lagunensis* populations were usually at a minimum during winter in the upper Laguna Madre and reached their maximum densities in spring or early summer when water temperatures exceed 25°C (Buskey *et al.*, 1996). In contrast, the related New England brown tide algal species *Aureococcus anophagefferans* shows a maximum growth rate at 20°C and 30 PSU (Cosper *et al.*, 1989).

There has been relatively little study of the effects of hypersalinity on zooplankton. In his study of the Alazan Bay region of Baffin Bay, Texas, Cornelius (1984) found the highest species diversity of zooplankton in waters ranging in salinity from 21 to 30 PSU, and the greatest abundance of zooplankton and largest number of taxa in waters of 31–40 PSU. Hypersaline waters (41–50 PSU) exhibit both the smallest number of taxa and the lowest average abundance of zooplankton (Cornelius, 1984). It seems likely from the results of field samples taken in this study that hypersaline conditions of 60 PSU and above have negative effects on microzooplankton populations. The reduction in abundance of microzooplankton grazers in hypersaline waters prior to the onset of the bloom is thought to have been a contributing factor in the initiation of this persistent bloom (Buskey *et al.*, 1997). Although the copepod *Acartia tonsa* does not appear to be an important grazer on *A.lagunensis* (Buskey and Stockwell, 1993; Buskey and Hyatt, 1995), hypersalinity also appears to have adverse effects on this copepod at salinities above 55 PSU (Figure 4) with hypersaline conditions correlated to a reduction in size of adult females. It is impossible to know whether this was a direct effect of salinity, or the result of a lower quality and/or quantity of food being available to *A.tonsa* at higher salinities before the onset of the bloom (e.g. the lower availability of microzooplankton; Figure 2). Interestingly, *A.tonsa* also showed a reduction in body size at all salinities after the initiation of the bloom when *A.lagunensis* dominated phytoplankton biomass and microzooplankton abundances were depressed (Buskey and Stockwell, 1993).

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The heterotrophic dinoflagellate *O.marina* is one of the protozoan species found to grow well on a diet of *A.lagunensis* alone (Buskey and Hyatt, 1995). It is a common protozoan in brown tide-affected areas (E.J.Buskey, unpublished data). *Oxyrrhis marina* has been reported to dominate in environments with high concentrations of phytoplankton, such as brackish tide pools, and as a contaminant in intensive algal cultures (Goldman *et al.*, 1989). In a study of the growth characteristics of an *O.marina* culture isolated from a tide pool in Finland, Droop (1959) found a maximum growth rate at a salinity of 16 PSU (compared to lower growth rates at 4, 8, 32 and 64 PSU) and optimum growth at a temperature of 22–23°C, with an upper tolerance limit of 28°C. For our culture of *O.marina* isolated in Texas coastal waters, we found somewhat different temperature and salinity tolerances, with maximum growth rates maintained at 30°C and 20 PSU (Figure 7). It seems likely that the adverse effects of high salinity on this and other protozoan grazers would have contributed to the persistence of the bloom of *A.lagunensis*, which can continue to grow at high salinities.

In the late summer and fall of 1997, extensive rainfall ended hypersaline conditions in the Laguna Madre and Baffin Bay, and salinities were lowered to <30 PSU. During this time, *A.lagunensis* populations fell from $>10^6$ cells ml^{-1} in Baffin Bay in July 1997 to <2000 cells ml^{-1} in October 1997 (E.J.Buskey, unpublished data), ending the >7 year uninterrupted bloom of *A.lagunensis* in Baffin Bay and the Laguna Madre. Although other factors besides reductions in salinity undoubtedly contributed to the demise of this persistent bloom, the results of this study and previous studies suggest that the grazers and competitors of *A.lagunensis* would be favored by this drop in salinity.

Acknowledgements

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References

Armstrong,N.E. (1987) The ecology of open-bay bottoms of Texas: a community profile. *US Fish. Wildl. Serv. Biol. Rep. 85* (7.12).

Brown,L.F.,Jr, McGowen,H.H., Evans,T.J., Groat,C.G. and Fisher,W.L. (1977) *Environmental Geologic Atlas of the Texas Coastal Zone—Kingsville Area.* Bureau of Economic Geology, University of Texas at Austin, 131 pp.

Buskey,E.J. and Hyatt,C.J. (1995) Effects of the Texas (USA) 'brown tide' alga on planktonic grazers. *Mar. Ecol. Prog. Ser.*, **126**, 285–292.

Buskey,E.J. and Stockwell,D.A. (1993) Effects of a persistent 'brown tide' on zooplankton populations in the Laguna Madre of south Texas. In Smayda,T.J. and Shimizu,Y. (eds), *Toxic Phytoplankton Blooms in the Sea. Proceedings of the 5th International Conference on Toxic Marine Phytoplankton.* Elsevier Science, Amsterdam, pp. 659–666.

Buskey,E.J., Stewart,S., Peterson,J. and Collumb,C. (1996) Current status and historical trends of

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brown tide and red tide phytoplankton blooms in the Corpus Christi Bay National Estuary Program study area. CCBNEP-07.

Buskey,E.J., Montagna,P.A., Amos,A.F. and Whitledge,T.E. (1997) Disruption of grazer populations as a contributing factor to the initiation of the Texas brown tide algal bloom. *Limnol. Oceanogr.*, **42**, 1215–1222.

Cornelius,S.E. (1984) An ecological survey of Alazan Bay, Texas. Technical Bulletin No. 5. Caesar Kleburg Wildlife Research Institute, 163 pp.

Cosper,E.M., Dennison,W.C., Milligan,A., Carpenter,E.J., Lee,C., Holzapfel,J. and Milanese,L. (1989) An examination of environmental factors important to initiating and sustaining brown tide blooms. In Cosper,E.M., Bricelj,V.M. and Carpenter,E.J. (eds), *Novel Phytoplankton Blooms. Coastal and Estuarine Studies 35.* Springer-Verlag, Berlin, pp. 317–340.

DeYoe,H.R. and Suttle,C.A. (1994) The inability of the Texas 'brown tide' alga to use nitrate and the role of nitrogen in the initiation of a persistent bloom of this organism. *J. Phycol.*, **30**, 800–806.

DeYoe,H.R., Stockwell,D.A., Bidagare,R.R., Latasa,M., Johnson,P.W., Hargraves,P.E. and Suttle,C.A. (1997) Description of the algal species *Aureoumbra lagunensis* gen. et sp. nov. and referral of *Aureoumbra* and *Aureococcus* to the Pelagophyceae. *J. Phycol.*, **33**, 1042–1048.

Droop,M.R. (1959) A note on some physical conditions for cultivating *Oxyrrhis marina. J. Mar. Biol. Assoc. UK*, **38**, 599–604.

Dunton,K.H. (1994) Seasonal growth and biomass of the subtropical seagrass *Halodule wrightii* in relation to continuous measurements of underwater irradiance. *Mar. Biol.*, **120**, 479–489.

Friedman,G.M. and Krumbein,W.E. (1985) *Hypersaline Ecosystems.* Springer-Verlag, Berlin, 484 pp.

Gifford,D.J. (1985) Laboratory culture of marine planktonic oligotrichs (Ciliophora, Oligotricha). *Mar. Ecol. Prog. Ser.*, **23**, 257–267.

Goldman,J.C., Dennett,M.R. and Gordin,H. (1989) Dynamics of herbivorous grazing by the heterotrophic dinoflagellate *Oxyrrhis marina. J. Plankton Res.*, **11**, 391–407.

Hedgepeth,J.W. (1967) Ecological aspects of the Laguna Madre, a hypersaline estuary. In Lauff,G.H. (ed.), *Estuaries. AAAS Publ. 83.* AAAS, Washington DC, pp. 408–419.

Hildebrand,H.H. and King,W.D. (1979) A biological study of the Cayo del Oso and the Pita Island area of the Laguna Madre, 1972–1978. Final Report. Texas A&I University and Central Power and Light Company, Kingsville, TX, 472 pp.

Kain,J.M. and Fogg,G.E. (1958) Studies on the growth of marine phytoplankton I. *Asterionella japonica* Gran. *J. Mar. Biol. Assoc. UK*, **37**, 397–413.

Kain,J.M. and Fogg,G.E. (1960) Studies on the growth of marine phytoplankton III. *Prorocentrum micans* Ehrenburg. *J. Mar. Biol. Assoc. UK*, **39**, 33–50.

Keller,M.D., Selvin,R.C., Claus,W. and Guillard,R.R.L. (1987) Media for the culture of oceanic ultraphytoplankton. *J. Phycol.*, **23**, 633–638.

Kottmeier,St.T. and Sullivan,C.W. (1988) Sea ice microbial communities (SIMCO). 9. Effects of temperature and salinity on rates of metabolism and growth of autotrophs and heterotrophs. *Polar Biol.*, **8**, 293–304.

Lopez-Barrerio,T., Villareal,T.A. and Morton,S.A. (1998) Development of an antibody against the Texas Brown Tide (*Aureoumbra lagunensis*). In Reguera,B., Blanco,J., Fernandez,M.L. and Wyatt,T. (eds), *Harmful Algae.* Xunta de Galicia and Intergovernmental Oceanographic Commission of UNESCO, Paris, pp. 359–362.

Montagna,P.A., Stockwell,D.A. and Kalke,R.D. (1993) Dwarf surfclam *Mulinea lateralis* (Say 1822) populations and feeding during the Texas brown tide. *J. Shellfish Res.*, **12**, 833–842.

Moreira,G.S. (1975) Studies on the salinity resistance of the copepod *Euterpina acutifrons* (Dana). In Vernberg,F.J. (ed.), *Physiological Ecology of Estuarine Organisms.* University of South Carolina Press, pp. 73–80.

Onuf,C.P. (1996) Seagrass responses to long-term light reduction by brown tide in upper Laguna Madre, Texas: distribution and biomass patterns. *Mar. Ecol. Prog. Ser.*, **138**, 219–231.

Quammen,M.L. and Onuf,C.P. (1993) Laguna Madre: seagrass changes continue decades after salinity reduction. *Estuaries*, **16**, 302–310.

Shormann,D.E. (1992) The effects of freshwater inflow and hydrography on the distribution of brown tide in South Texas. MA Thesis, Department of Marine Science, University of Texas at Austin, 112 pp.

Stockwell,D.A., Buskey,E.J. and Whitledge,T.E. (1993) Studies of conditions conducive to the development and maintenance of a persistent 'brown tide' in Laguna Madre, Texas. In Smayda,T.J. and Shimizu,Y. (eds), *Toxic Phytoplankton Blooms in the Sea. Proceedings of the 5th International Conference on Toxic Marine Phytoplankton.* Elsevier Science, Amsterdam, pp. 693–698.

Stoecker,D.K., Gustafson,D.E., Merrell,J.E., Black,M.M.D. and Baier,C.T. (1997) Excystment and

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growth of chrysophytes and dinoflagellates at low temperatures and high salinities in Antarctic sea-ice. *J. Phycol.*, **33**, 585–595.

Vernberg,F.J. and Vernberg,W.B. (1975) Adaptations to extreme environments. In Vernberg,F.J. (ed.), *Physiological Ecology of Estuarine Organisms*. University of South Carolina Press, pp. 165–180.

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